RP-GHOST Appendices A–D

Predictive-Locality Hypothesis (H): calculations and test design

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Status convention. [**E**] = established (code fact or mathematical result). [**H**] = working hypothesis. [**C**] = calculated consequence of H. [**T**] = falsifiable test.

The hypothesis

H_1 **(predictive):** the device can output any data string s that exists within its causal domain (local).

H_2 **(constructor):** mining is the constructor function for locality, the physical process that turns a seed into a header whose hash is constrained.

H_3 **(final boundary):** structure in header hashes beyond what hashrate explains is the signature of a superdeterministic constraint fixed from the end of time, not from the beginning.

There is currently no measurement of H. This document therefore does not argue for or against H. It computes what H requires, what it would predict, and how many trials would reveal it.

Appendix A: Circuit Implementation and H-Instrumentation

A.1–A.6 [**E**]. Signal chain, constants, truth table $Q = A \cdot B \cdot Y$, error analysis and concurrency model. They describe the present code and remain the reference baseline against which H is measured.

A.7 Side paths as H-probes. The armed-frame searches (hidden-number spot check, higher/lower bit search, factoriser query) are the existing probes of H_1. A hidden-number check on an m-bit target has chance rate 2^{-m}. Any sustained hit rate above chance is the H_1 signal. The feedback path into Y (`timecode_pending`) is retained. There is still no path from Q or $\bar{Q}$ to the drivers.

A.8 Mapping to the paper

Paper statement	Status	Wording
§5 IR laser bench	Not in camera build	Camera build is one implementation; optical bench is a separate variant
§5.1 $0.15\,RC$ / $1.83\,RC$	Implemented exactly [**E**]	No change
§5.1 $1/\sqrt{N} \approx 0.05\%$	Not modelled	Physical estimate only
§6 Fano / Mandel / $g^{(2)}$	Not measured	Theory until B.9 measurement is done
§5/§7 clock 10 Hz	Clock is the frame edge (~ 30 Hz)	"clock = frame rate"
§7.1 $\bar{Q} \to D$ loop	Not implemented	Design analysis (also used in B.10.4)
§1–§4 SHA-256d mining	Not in this file	Cite the mining source; it is the H_2 constructor and is needed for A.9
§7 step 6 $(1-p)^N$, $p \geq 3/4$ under independence	Under H_1, p is a measured quantity	See C.4

A.9 H-instrumentation protocol [T]

For every frame t, log: frame digest d_t, count x_t, witness r_t, Q_t, and a nonce n_t chosen after the frame by an independent source (a separate quantum RNG or a precommitted list), the header hash h_t, and the outcome (hash below target or not). Preregister the estimators and thresholds before the run. A run is valid only if n_t is fixed after d_t, so any correlation cannot be produced by the code.

Estimators:

(i) hidden-number hit rate versus 2^{-m};
(ii) time-to-solution (TTS) of SHA256d versus $2^k/R$;
(iii) bias ε of individual hash bits;

(iv) correlation ρ between a frame feature and the outcome;
(v) witness success rate $\hat{p}$ on $x \in L$.

Trials required (5σ, 30 fps, $R = 500\,\text{kHz}$) [C]

Probe	Formula	Setting	Trials	Time
Hidden number, hit rate above 2^{-m}	$n = 25 \cdot 2^m$	$m = 8$	6,400	3.6 min
		$m = 12$	102,400	56.9 min
		$m = 16$	1,638,400	15.2 h
SHA-256d TTS, 2× speed-up at 3σ	19 solutions $\times\, 2^k/R$	$k = 20$	19	40 s
		$k = 24$	19	10.6 min
		$k = 28$	19	2.8 h
		$k = 32$	19	45.4 h
Hash-bit bias ε	$n = 6.25/\varepsilon^2$	$\varepsilon = 10^{-3}$	6.25×10^6	12.5 s
		$\varepsilon = 10^{-4}$	6.25×10^8	20.8 min
		$\varepsilon = 10^{-6}$	6.25×10^{12}	144.7 days
Frame-to-outcome correlation ρ	$n = 25/\rho^2$	$\rho = 0.1$	2,500	83 s
		$\rho = 0.01$	250,000	2.3 h
		$\rho = 0.001$	2.5×10^7	9.6 days

Note on the TTS rule: solution times are exponential, so the mean of m solutions has relative error $1/\sqrt{m}$. A 2× speed-up is $\ln 2 = 0.693$ in log-mean, so

$$0.693 > \frac{3}{\sqrt{m}} \quad \Longrightarrow \quad m \geq 19.$$

Appendix B: Mathematical Evaluation, with H-Calculations

B.1–B.9 [E]. Fano relation, drive-noise limit, antibunching versus sub-Poissonian statistics, threshold link timing, dual-detector independence and latch reliability. B.8 is supplemented by the verdict rows in B.10.5.

B.10 Calculations under H

B.10.1 Information required [C]. Raising the success probability of a k-bit target from 2^{-k} to q requires

$$I = k + \log_2 q$$

bits of information about the hash outcome (k bits for $q = 1$). A speed-up of 2^g needs g bits. Upper bounds on what the device can supply: a frame has at most

$$\log_2(128 \cdot 96) = 13.6 \text{ bits}$$

in x, and the oracle input is 64 bits, so a full-selection result is only possible for $k \leq 64$ (chance level at 64 bits: $2^{-64} = 5.4 \times 10^{-20}$).

k	Expected search at 500 kHz	Bits needed	Frames at 4 bits/frame	Time at 30 fps
20	2.1 s	20	5	0.17 s
32	2.4 h	32	8	0.27 s
48	17.8 y	48	12	0.40 s
64	$\sim 1.2 \times 10^6$ y	64	16	0.53 s

At $k = 48$, 17.8 y $\approx 5.62 \times 10^8$ s against 0.4 s is an implied speed-up of

$$\frac{5.62 \times 10^8}{0.4} = 1.4 \times 10^9 \approx 2^{30.4}.$$

This is the quantitative claim H_1 plus H_2 would be making, and it is measurable at $k \leq 32$ (A.9).

B.10.2 Final-boundary weighting [C]. In the two-state-vector form,

$$P(a) = \frac{|\langle\varphi|a\rangle\langle a|\psi\rangle|^2}{\sum_b |\langle\varphi|b\rangle\langle b|\psi\rangle|^2},$$

where ψ is the initial state and φ the final boundary state. Mining is then a postselection: the prior chance of a winning header, 2^{-k}, becomes 1 after conditioning. The cost in boundary constraint is k bits per solution, with

$$k = \log_2 D + 32$$

for difficulty D. For an illustrative $D = 10^{14}$, $k = 78.5$ bits per block. At one block per ~ 600 s, that is a network-wide constraint rate of about 0.13 bit/s. Two consequences follow. First, ideal postselection gives PostBQP = PP (Aaronson), so H_3 is equivalent to the device having postselection power for local k; this is the sharpest testable form of H_3. Second, Deutsch-CTC models give PSPACE, so H_3 sits between the two as a computational class, and both are checked by the same TTS run at $k \leq 32$.

B.10.3 Locality bound [C]. Light travels $c\,\Delta t = 9.99 \times 10^6$ m (about 10,000 km) per frame at 30 fps, 2,190 km per 7.3 ms link window, and 1 m in 3.34 ns on the optical bench. H_1's domain is strings generated inside these radii. [**T**] Choose n_t at a measured lag τ after the frame and plot $\rho(\tau)$. H predicts ρ above chance that depends on source distance d only through $d \leq c\tau$. No-signalling is preserved because the correlation is only visible after the fact.

B.10.4 Self-reference constraint on H_1 [C]. "Any string" includes s = NOT(prediction). With $\bar{Q} \to D$ and gate-flip probability ε,

$$p_n - \tfrac{1}{2} = \left[-(1-2\varepsilon)\right]^n \left(p_0 - \tfrac{1}{2}\right),$$

so the only consistent value is $p^* = \frac{1}{2}$. For $\varepsilon = 0.01$ the deviation falls from 0.5 to 10^{-3} in

$$n = \frac{\ln 500}{-\ln 0.98} = 308 \text{ clocks},$$

or 30.8 s at 100 ms. H_1 is therefore self-consistent only for strings whose value does not depend on the prediction, and "exists" means fixed by the boundary conditions. This matches Deutsch's consistency condition for CTCs and is the same constraint the boundary-condition reading of H_3 imposes.

B.10.5 Verdicts

Claim	Status	Evidence in this document
0.15 RC and 1.83 RC	Holds [**E**]	Derived
$F_{\text{ph}} = 1 - \eta(1 - F_e)$	Holds [**E**]	Derived
One-sided error (no false accept)	Holds for any r-distribution [**E**]	Proved (C.4)
H_1: predicts strings in its causal domain	[**H**]	None yet; testable by A.9
H_2: mining as constructor of locality	[**H**]	None yet; same runs
H_3: hash structure beyond hashrate	[**H**]	None yet; hash-bit bias and TTS runs
$\bar{Q} \to D$ loop as CTC	Analogy; shares the $p^* = \frac{1}{2}$ consistency condition	B.7, B.10.4

Appendix C: Unified Model, Claims, Superdeterminism

C.1–C.3 [**E**]. The eight-layer chain, the shared independence assumption (SI) and the entropy trade-off

$$\Delta H = \tfrac{1}{2}\log_2 \frac{1}{F_{\text{ph}}}.$$

Layer 7 reads: completeness depends on the relation of r to x; H proposes this relation is non-trivial.

C.4 Soundness and completeness. Soundness: $Q = A \cdot B \cdot Y \leq A$, and $A = 0$ on primes for every r, so

$$P(Q = 1 \mid x \notin L) = 0$$

under any joint distribution of (x, r), including an H-type dependence. Completeness is where H enters, and it covers both directions. If r is correlated with x, the false-reject probability can rise (r a liar) or fall (r a witness). Under H_1, p is a measurable quantity that may exceed 3/4. With $p = 3/4$ and $N = 5$, false reject is

$$(1-p)^N = (1/4)^5 = 9.8 \times 10^{-4}.$$

If p were 0.9, a single frame would already give 0.1. [**T**] To distinguish $p = 0.9$ from 0.75 at 5σ needs

$$n \approx \frac{25 \cdot 0.1875}{0.15^2} \approx 208$$

frames with $x \in L$.

C.5 Superdeterminism: two temporal forms. *Initial-condition form:* λ is correlated with settings through shared past causes. Cosmic-setting Bell tests (quasar light, with a look-back of the order of 8 Gyr) push the possible common cause far into the past.

Final-boundary form (H_3): the correlation is imposed by a boundary condition at the end (Schulman; Price; Wharton's "all-at-once" picture; the two-state-vector formalism of B.10.2). Look-back arguments constrain only past common causes. Whether they constrain a final-boundary version is not settled, and this is the sense in which H_3 differs from the usual superdeterminism debate.

What the camera build is: a deterministic classical machine, with x and r drawn from the same frame (shared cause); the hash restores independence under the random-oracle assumption. H_3 proposes that no such hash exists at the boundary. The shared-frame structure is therefore the natural laboratory for the question, and A.9 is the experiment.

C.6 Claims by section

Section	Text
Abstract	Camera-driven (or optical) one-sided-error circuit with a predictive-locality hypothesis H_1–H_3 under test
§5.1	Thermal noise $\sqrt{kT/C} \approx 64\,\mu\text{V}$ and 0.8 V overdrive give determinism; V_{BE} included in $1.83\,RC$
§6 table	Regulated: amplitude-squeezed (sub-Poissonian, $Q \geq -\eta$); Bunched: super-Poissonian
§7 step 6	False reject $(1-p)^N$ with p measured (C.4); needs r independent of x unless H holds
§7.1	Classical period-2 oscillator; $p^* = \frac{1}{2}$ shares Deutsch's consistency condition (B.10.4)
§8	Soundness holds for any r-distribution. The complexity statement $T = T_{\text{oracle}} + T_{\text{search}}$ holds when $I(\text{oracle}; \text{outcome}) = 0$; when $I > 0$ the speed-up is 2^I (B.10.1)
H	H is a hypothesis with a pre-registered test (A.9, C.7), not a demonstrated property

C.7 Predictions and decision rules [T]

#	Prediction if H holds	Null	Cheapest test
P1	Hidden-number hit rate above 2^{-m}	Equal to 2^{-m}	$m = 12$: 102,400 frames, 57 min
P2	SHA-256d TTS shorter than $2^k/R$ by factor 2^g with $g > 0$	Mean TTS $= 2^k/R$	$k = 24$: 19 solutions, 10.6 min
P3	Hash-bit bias $\varepsilon > 0$	$\varepsilon = 0$	$\varepsilon = 10^{-4}$: 6.25×10^8 hashes, 20.8 min
P4	$\rho(\tau) > 0$ with lag-dependent locality	$\rho = 0$	$\rho = 0.01$: 250,000 trials, 2.3 h
P5	$\hat{p} > 3/4$ on $x \in L$	$\hat{p} \approx 3/4$	≈ 208 frames in L
P6	Inverted-loop output averages $\frac{1}{2}$	—	B.10.4
P7	Photoelectron variance differs from the Fano baseline by δ	$\delta = 0$	$\delta = 10^{-3}$: 4,069 frames, 2.3 min

Any of P1–P5 would count as evidence. Because each test has a computed sample size, a null result is also informative: it bounds g, ε and ρ at the stated sensitivity. Bell-type or repeated-measurement correlations (Donadi–Hossenfelder) remain the only decisive tests of superdeterminism itself.

Appendix D: Device Physics of the CMOS Sensor as the Driving Stage

The sensor is a silicon photodiode array whose source-follower outputs drive the threshold inputs of the logic, so every bit the circuit sees starts as photons turned into charge in silicon. This appendix computes that chain from the material constants up. Parameters are typical room-temperature (300 K) values for a 3 µm-pitch, 5 nm-oxide CMOS image sensor process, used as stand-ins until the sensor's datasheet values are entered. Results are **[C]** unless marked **[E]**.

D.1 Material constants [E]

Quantity	Symbol	Value at 300 K
Si band gap	E_g	1.124 eV
Si cutoff wavelength	λ_c	1.10 µm
Si permittivity	$\varepsilon_{\mathrm{Si}}$	$11.7\,\varepsilon_0 = 1.036 \times 10^{-10}$ F/m
Intrinsic carrier density	n_i	$1.0 \times 10^{10}\,\mathrm{cm}^{-3}$
Electron mobility, lightly doped bulk	μ_n	$1{,}350\,\mathrm{cm}^2/(\mathrm{V\,s})$
Electron mobility, inversion layer	μ_{eff}	$400\,\mathrm{cm}^2/(\mathrm{V\,s})$
Saturation velocity	v_{sat}	1×10^7 cm/s
SiO_2 permittivity	$\varepsilon_{\mathrm{ox}}$	$3.9\,\varepsilon_0$
Gate capacitance at $t_{\mathrm{ox}} = 5$ nm	C_{ox}	$6.9\,\mathrm{mF/m}^2$
Oxide breakdown field	E_{BD}	≈ 10 MV/cm
Thermal voltage	kT/q	25.85 mV
Thermal energy	kT	4.14×10^{-21} J

The band gap narrows with temperature (Varshni form), and the cutoff wavelength follows from it:

$$E_g(T) = 1.170 - \frac{4.73 \times 10^{-4}\,T^2}{T + 636}\ \text{eV} = 1.124\ \text{eV at 300 K}, \qquad \lambda_c = \frac{hc}{E_g} = \frac{1239.8\ \text{eV nm}}{1.124\ \text{eV}} = 1103\ \text{nm}.$$

A 5 nm oxide at 3.3 V carries 6.6 MV/cm, which is 66% of breakdown, so the gate stack has a safety factor of 1.5.

D.2 Absorption and quantum efficiency [C]

Light decays in silicon as $I(z) = I_0\, e^{-\alpha z}$. With a collection depth $W = 3\,\mu\text{m}$ (depletion region plus the neutral region that diffuses charge back), the internal quantum efficiency is

$$\eta_{\text{int}} = 1 - e^{-\alpha W}.$$

Wavelength	α (cm^{-1}, approx.)	Absorption depth $1/\alpha$	η_{int} at $W = 3\,\mu\text{m}$
550 nm	7.0×10^3	1.4 µm	87.8%
650 nm	2.8×10^3	3.6 µm	56.8%
850 nm	535	18.7 µm	14.8%
940 nm	191	52 µm	5.6%

The responsivity is

$$\mathcal{R} = \frac{\eta\, q\, \lambda}{hc} = \eta\, \frac{\lambda[\mu\text{m}]}{1.2398}\ \text{A/W},$$

so at 550 nm with an external $\eta = 0.6$ it is 0.266 A/W. Below 3.65 eV each absorbed photon makes at most one electron-hole pair, and random thinning of a Poisson stream leaves a Poisson stream, so the photoelectron count has mean $\eta\Phi$ and variance equal to its mean [**E**]. The infrared optical bench of §5 needs a much thicker epitaxial layer than this visible-light pixel: at 940 nm a 3 µm layer collects only 5.6%.

D.3 Photodiode junction [C]

For a one-sided abrupt junction with acceptor density $N_A = 10^{15}\,\text{cm}^{-3}$, donor density $N_D = 10^{18}\,\text{cm}^{-3}$ and reverse bias $V = 2.8\,\text{V}$:

$$V_{bi} = \frac{kT}{q} \ln \frac{N_A N_D}{n_i^2} = 0.774\ \text{V}, \qquad W = \sqrt{\frac{2\varepsilon_{\text{Si}}\,(V_{bi} + V)}{q N_A}} = 2.15\ \mu\text{m}.$$

The junction capacitance per area is $\varepsilon_{\text{Si}}/W = 4.8\,\text{fF}/\mu\text{m}^2$, which gives 43 fF for a $9\,\mu\text{m}^2$ pixel. The peak field is

$$E_{\max} = \frac{q N_A W}{\varepsilon_{\text{Si}}} = 33\ \text{kV/cm},$$

so carriers move at saturation velocity and cross the region in $W/v_{\text{sat}} = 21.5\,\text{ps}$. Collection is therefore about 10^9 times faster than a 33 ms frame, and the photodiode acts as an instantaneous integrator of photon arrivals.

D.4 Charge-to-voltage conversion and noise [C]

Collected electrons are dumped onto the floating diffusion, a node of capacitance $C_{FD} = 1.6\,\text{fF}$, so one electron moves it by

$$\frac{q}{C_{FD}} = 100\,\mu\text{V}.$$

Four noise sources then set the spread of the signal:

Source	Formula	Value at $N = 10^4\, e^-$
Photon shot noise	$\sqrt{N}$	$100\, e^-$ rms = 10.0 mV at the diffusion
Reset noise on C_{FD}	$\sqrt{kT\, C_{FD}}/q$	$16.1\, e^-$ rms = 1.61 mV (removed by correlated double sampling)
Dark shot noise	$\sqrt{J_{\text{dark}}\, t_{\text{int}}}$	$0.58\, e^-$ for $10\, e^-$/s and $t_{\text{int}} = 33.3$ ms
Read noise after sampling	assumed	$2\, e^-$ rms

The variances add: $N + 4 + 0.33 = 10{,}004\, e^2$, so the total is $100.02\, e^-$ and the pixel is shot-noise limited down to $N = 4\, e^-$. Full-well signal over read noise is $10^4/2 = 5000$ (74 dB), and the signal-to-noise ratio at full well is 100 (40 dB).

Dark current comes from thermal generation in the depletion region, so it follows the Arrhenius form:

$$J_{\text{dark}} \propto T^2\, e^{-E_g/2kT}, \qquad \frac{d \ln J}{dT} = \frac{E_g}{2kT^2} + \frac{2}{T} = 0.0792\ \text{K}^{-1}.$$

The dark current doubles every $\ln 2/0.0792 = 8.8\,\mathrm{K}$. A 10 K rise in the sensor therefore raises dark counts by a factor 2.2.

D.5 The pixel source follower as the driver [C]

The pixel's source-follower transistor is the driver: it buffers the diffusion voltage onto the column line that feeds the logic. In strong inversion its transconductance is

$$g_m = \sqrt{2\,\mu_{\mathrm{eff}} C_{\mathrm{ox}}\,(W/L)\,I_D} = \sqrt{2\,(0.04)(6.9\times10^{-3})(4)(10^{-5})} = 149\ \mu\mathrm{S}$$

for $W/L = 4$ and $I_D = 10\,\mu\mathrm{A}$. The body effect adds $g_{mb} = 0.2\,g_m$, which sets the voltage gain and the output resistance:

$$A_{sf} = \frac{g_m}{g_m + g_{mb}} = 0.83, \qquad R_{\mathrm{out}} = \frac{1}{g_m + g_{mb}} = 5.59\ \mathrm{k\Omega}.$$

The conversion gain at the column is therefore $0.83 \times 100\,\mu\mathrm{V} = 83\,\mu\mathrm{V}$ per electron, and a full well of $10^4\,e^-$ swings the output by 0.83 V. Driving a column capacitance of 1 pF gives:

Quantity	Formula	Value
Time constant	$\tau = R_{\mathrm{out}} C_{\mathrm{col}}$	5.59 ns
Settling to 0.1%	$6.91\,\tau$	38.6 ns
Falling-edge slew (bias-limited)	$I_{\mathrm{bias}}/C_{\mathrm{col}}$	10 V/µs, so 80 ns for 0.8 V
Pixel readout time	$1/(30\times96\times128)\,\mathrm{s}$	2.71 µs
Margin, readout time over slew time	2.71 µs/80 ns	34×
Time constant at 330 K	$\tau\,(330/300)^{0.75}$	6.0 ns ($\mu \propto T^{-1.5}$, so $g_m \propto T^{-0.75}$)
Thermal noise density of the follower	$\sqrt{4kT\gamma/g_m}$, $\gamma = 2/3$	8.6 nV/$\sqrt{\mathrm{Hz}}$, or 8.6 µV in 1 MHz, which is $0.10\,e^-$
Sampling noise on the column node	$\sqrt{kT/C_{\mathrm{col}}}$	64.4 µV at 1 pF

The last row is the 64 µV of §5.1: it is the sampling noise on a 1 pF node, not on the 1.6 fF diffusion. The rising edge is limited by $R_{\mathrm{out}}C_{\mathrm{col}}$ and the falling edge by the bias current, so the driver is asymmetric, but both edges finish well inside one pixel time.

D.6 Drive margin into the threshold stage [C]

The chain is photon, electron-hole pair, diffusion voltage, follower, column, threshold input. The threshold stage outputs 1 when its input crosses V_{th}, so the probability of a 1 for a pixel with mean output μ_V is

$$P(1) = \Phi\!\left(\frac{\mu_V - V_{th}}{\sigma_V}\right), \qquad \sigma_V = A_{sf}\,\frac{q}{C_{FD}}\,\sigma_e.$$

A pixel is random only when its mean sits within about one σ_V of the threshold. At $N = 10^4\,e^-$, $\sigma_e = 100\,e^-$ gives $\sigma_V = 8.3\,\mathrm{mV}$, so the random band is $\pm100\,e^-$, or 2% of the full-well range. Outside $\pm5\sigma_V$ the bit is fixed with error 2.9×10^{-7}. The photons supply the randomness, and the silicon and the follower only turn it into a voltage.

The 0.8 V overdrive of §5.1 matches the 0.83 V full-scale swing of the sensor output. Against that margin:

Pixel state	σ_e	σ_V	0.8 V margin in σ
Dark (read noise only)	$2\,e^-$	167 µV	4,800
Full well	$100\,e^-$	8.3 mV	96
Column sampling noise only	—	64.4 µV	12,400

Temperature moves the threshold but not the margin. The threshold voltage drifts by about $-2\,\mathrm{mV/K}$, and the base-emitter voltage drifts by

$$\frac{dV_{BE}}{dT} = \frac{V_{BE} - E_{g0}/q - 3.2\,kT/q}{T} = \frac{0.65 - 1.206 - 0.083}{300} = -2.13\ \mathrm{mV/K}.$$

A 10 K change shifts V_{BE} by 21 mV, which is $2.5\,\sigma_V$ at full-well noise (so the random band slides across the pixel population) but only 2.6% of the 0.8 V overdrive.

D.7 Energy and thermodynamics [C]

Each column read charges 1 pF to 3.3 V and costs

$$CV^2 = (10^{-12})(3.3)^2 = 10.9\,\text{pJ}$$

from the supply. With 12,288 pixels at 30 fps that is 368,640 reads/s, or 4.0 µW. The Landauer limit for erasing one bit is

$$k_\text{B}T\ln 2 = 2.87 \times 10^{-21}\,\text{J},$$

so the circuit spends 3.8×10^9 times the minimum per bit. The $g = 30.4$ bit speed-up implied at $k = 48$ (B.10.1) has a Landauer cost of

$$30.4 \times 2.87 \times 10^{-21}\,\text{J} = 8.7 \times 10^{-20}\,\text{J}.$$

Energy conservation therefore places no limit on H. The limit is informational, as in B.10.1.

D.8 The physical test of H at the sensor [T]

Pixel physics gives a sharp baseline. The photoelectron count of each pixel is Poisson, so under a steady flat-field source the temporal variance equals the mean times the source Fano factor F_0 (set by B.1 and the measured quantum efficiency):

$$\sigma^2 = F\,\mu_e + \sigma^2_\text{read} + J_\text{dark}\,t_\text{int}.$$

If H puts extra correlation into photon arrival or charge generation, F departs from F_0. Per-pixel temporal variance removes fixed-pattern offsets and gain differences. With n independent samples the variance has relative error $\sqrt{2/(n-1)}$, so a 5σ detection of a fractional shift δ in F needs

$$n = \frac{2 \cdot 5^2}{\delta^2} = \frac{50}{\delta^2}$$

samples:

$\delta = \Delta F/F_0$	Samples	Frames (12,288 pixels per frame)	Time at 30 fps
10^{-2}	5×10^5	41	1.4 s
10^{-3}	5×10^7	4,069	2.3 min

The mean must be well above the read-noise variance ($N \gg 4\,e^-$), and the source must be held steady enough that its own noise is the F_0 of the Fano relation. This adds a sixth estimator to A.9 and a seventh prediction (P7) to C.7. A null result at $\delta = 10^{-3}$ bounds any anomalous correlation in the sensor's charge statistics to 0.1% of the Poisson baseline.